

RESEARCH ARTICLE

Postrelease Dispersal of Reintroduced Elk (*Cervus elaphus*) in Ontario, Canada

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Abstract

We studied 2 years of postrelease telemetry data of elk (*Cervus elaphus*) translocated to their historic range limit in Ontario, Canada and sought to determine if postrelease movements were related to behavior, demography of released animals, or site-specific attributes such as length of holding period. During 1998–2004 we radio-tracked 341 elk in 10 release groups via ground and aerial telemetry and monitored movement patterns relative to gender, age, and pre-release holding period (4–112 days). We found that elk that were held for short periods prior to release (4–11 days) moved longer distances than those subject to extended conditioning (17–112 days), suggesting that an extended conditioning period is beneficial from the standpoint of promoting philopatry. When all elk were

pooled by sex and age class, male calves remained in closer proximity (8.0 ± 13.2 km) to release sites than adult females (19.1 ± 20.6 km), adult males (19.7 ± 15.1 km), and female calves (14.4 ± 20.4 km). Most calves dispersed in a southeasterly direction whereas adults tended to travel southwest. Our results reveal that elk movement characteristics are influenced by factors such as release protocol and group demographics; these findings provide further insight regarding appropriate release methods for restoring natural populations near their historical range limit.

Key words: *Cervus elaphus*, dispersal, elk, movement, Ontario, reintroduction, telemetry.

Introduction

Reintroduction, defined as the release of animals to areas of historical occupation (IUCN 2003), has become a popular tool in wildlife management and conservation biology (Griffith et al. 1989; Kleiman 1989; Fischer & Lindenmayer 2000; Calvete & Estrada 2004). Griffith et al. (1989) reported that over 700 translocations were conducted annually in North America during 1973–1986, and translocated species included a variety of mammals, birds, and reptiles, with almost half (44%) focusing on threatened, endangered, or vulnerable species. Griffith et al.'s (1989) analysis revealed that success increased with high habitat quality, among herbivores, and when released animals received special protection status. Probability of success also increased when animals were reintroduced to the core of the species' historic range, in areas lacking competitors, and when wild-caught animals were released (Griffith et al. 1989; Wolf et al. 1996; Jule et al. 2008).

Restoration programs typically are undertaken in areas where populations have faced marked constraints, and for many species such areas often are at the periphery of the range where habitat quality is marginal and population regulation mechanisms are substantive. However, factors determining release success at the edge of the range remain poorly understood for many species, including most ungulates. While reintroduction is doubtless an important tool for wildlife management and conservation, information remains scant on the appropriate protocol and conditions for reintroduction of ungulates in areas where they have been extirpated near their historical range limit. Indeed, understanding such relationships may be critical to the success of ungulate reintroductions worldwide.

Elk or wapiti (*Cervus elaphus*) were indigenous and relatively abundant in Ontario at the time of European settlement (Bosveld 1996). Historic evidence indicates that Ontario elk occupied the deciduous forest to the west and east of Lake Superior, encompassing roughly 20–30% of the current area in the province (Fig. 1; Bellhouse & Rosatte 2005). However, by the early 1800s a number of factors, including overharvest, habitat destruction, and perhaps meningeal worm (*Parelaphostrongylus tenuis*) infection had caused the extirpation of the species in Ontario and eastern Canada (Ranta et al. 1982; Bellhouse & Broadfoot 1998; Bellhouse & Rosatte 2005). Restoration of elk to their historic range has occurred in many North American jurisdictions including Michigan, Wisconsin

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Figure 1. ERAs and approximate historical elk distribution (shaded area) in Ontario, Canada.

(Anderson et al. 2005b), Kentucky (Larkin et al. 2003), Arizona (Hicks & Rachlow 2005), Pennsylvania (Williams et al. 2002), and Arkansas (Telesco et al. 2007). Efforts continue to monitor and evaluate the success of these restoration programs. Restoration to Ontario presented a unique challenge because of the harsh climate and marginal habitat (Bellhouse & Broadfoot 1998). Since the distribution of dispersal distances greatly influences the rate of population spread and ultimately reproduction (Rousset & Gandon 2002; Tweed et al. 2003; Anderson et al. 2005a), release-site fidelity of elk should be an important determinant of translocation success.

The objective of the present study was to examine the postrelease movement of elk released in sites at the northern periphery of the species' historical distribution, and to compare efficacy of various release protocols on release success. Because male elk tend to naturally disperse longer distances and at a higher frequency than females (Clutton-Brock et al. 1982; Boyce 1989; de Vergie 1989), we speculated that similar patterns would characterize reintroduced elk postrelease. However, it is unclear if elk habitat at the range periphery is sufficiently poor to over-ride any gender- or individual-based differences in movement patterns (Anderson et al. 2005a). In addition, because the success of translocation programs is related to habitat quality in the release area (Griffith et al. 1989; Wolf et al. 1996), the marginal habitat in all four release areas relative to the core of the species' range lead us to predict that release protocols (e.g., longer pre-release conditioning and/or higher habitat quality in areas immediately surrounding a given release site) would limit postrelease movements.

Methods

Study Area

Between 1998 and 2001 eastern elk were released at four sites in Ontario; previously, these four sites had been

selected based on low deer (*Odocoileus virginianus*), moose (*Alces alces*), and wolf (*Canis lupus*) densities, low human population density and road density, and high availability of appropriate seasonal habitat (Bellhouse & Broadfoot 1998; Rosatte et al. 2007). The four Ontario release sites (Fig. 1) were Bancroft/North Hastings (BNH), Nipissing/French River (NFR), Lake Huron North Shore (LHNS), and Lake of the Woods (LOW). Latitudinal differences in release site locations resulted in marked habitat differences between sites. BNH was comprised of mixed deciduous and coniferous including various combinations of maple (*Acer saccharum*), spruce (*Picea glauca*), fir (*Abies balsamea*), aspen (*Populus tremuloides*), and birch (*Betula papyrifera*). The area surrounding the release site was characterized by low human density and sparse farmland (Yott 2005). NFR was characterized by mixed hardwood-conifer habitats subject to substantive human disturbance, including logging, mining, and fire (Chambers et al. 1996). LHNS was composed of a mix of coniferous and deciduous forests, and large tracts of agricultural land near the central portion of the area. LOW was located in a transition zone between the Great Lakes—St. Lawrence and boreal forest regions in northwestern Ontario. The area was composed of coniferous forests to the north and mixed coniferous/deciduous forests and agricultural areas to the south (Rowe 1972). Climatically, the study areas differed mainly in the amount of annual precipitation received, which varied between 250 and 1,000 mm following roughly a gradient from south to north. In all study areas, January and July were the coldest and warmest months, respectively (Anonymous 2003).

Data Collection and Preparation

The source of animals for Ontario's elk restoration program was Elk Island National Park (EINP) in Alberta, Canada. Prior to shipment, animals were treated with Fascinex (10% Triclabendazole) and Ivermectin (Ivomec, Merial Inc., USA) (10 mg/50 kg body weight) to control various nematodes and ectoparasites (Rosatte et al. 2002). Vitamin E and selenium also were provided to reduce potential capture myopathy. Elk were weighed, sexed, aged, and fitted with ear tags and mortality-sensitive VHF or GPS radio-collars (Lotek Engineering Inc., Newmarket, Ontario; Rosatte et al. 2002). Elk were transported to Ontario via cattle truck, and shipment time from EINP to Ontario varied between 24 hours (LOW) and 58 hours (BNH) (Rosatte et al. 2007). Upon arrival in each release area, elk were released into temporary holding facilities until spring green-up, with holding periods ranging from 4 to 112 days. For the purpose of this study, we define a semi-soft release as that occurring 4–11 days after placement in the holding pen, while a soft release occurred after a holding period of 17–112 days; this classification resulted in 85 and 256 animals being released in semi-soft and soft releases, respectively. Hard releases, in which animals were released onto the landscape immediately, were not considered as an option because we wanted them to remain in the areas defined as suitable by the habitat model of Bellhouse and Broadfoot (1998). During captivity, elk were provided with hay and water.

Following release, animals were located 1–5 times per week (typically during daylight hours) using standard ground and aerial telemetry protocols. Location estimates were calculated from a minimum of three bearings, and estimates with large errors (>350 m) were excluded from analysis. Elk that died or lost their radio-collars before the first rut of the year in which they were released (September 15; $n = 102$) were censored from the analysis. We used 2-years postrelease as our cut-off for assessing elk postrelease movement (dispersal) patterns because this was considered as the point at which animals should have dispersed and established a home range. For the purpose of this study, we define “dispersal” as linear movement between the release site and each telemetry location estimate—in the present analysis we draw no distinction between animals that disperse and reproduce (genetic dispersal) versus those that disperse but do not reproduce (ecological dispersal).

We calculated mean linear dispersal distance and mean dispersal bearings from release sites for each animal (White & Garrott 1990). To assess levels of independence we calculated coefficients of variation for each variable to determine the amount of variation between and within groups. Results indicated that sample variation was greater within release groups than between groups, implying that individual animals were appropriate as the experimental unit. Observations were stratified by release area, release group, sex, and age class (calves < 1 year at time of release; adults ≥ 1 year at time of release). Because these classes were mutually exclusive, we used single-factor ANOVAs (analysis of variance) to test for significant relationships in mean dispersal distance. We further explored significant differences using Tukey's posthoc test (Sokal and Rohlf 1996). Linear statistics were conducted in SPSS (Version 10, SPSS Inc., Chicago, Illinois).

Uniformity of dispersal direction was tested for each release group and release site using Watson's U^2 -test (Zar 1999). Concentration of dispersal bearings was also calculated. Concentration can vary from 0, when data is so dispersed that a mean angle cannot be described, to 1.0, when all the data are concentrated in the same direction (Zar 1999). Posthoc comparisons of dispersal bearing were further explored using appropriate multisample or pairwise Watson–Williams F-tests (Zar 1999). Circular distribution statistics were calculated in Oriana for Windows (Version 2.0, Kovach Computing Services), with $p < 0.05$ as significance level.

Table 2. Age- and sex-specific dispersal characteristics of elk.

	Males		Females	
	Adults	Calves	Adults	Calves
Mean distance (km $\pm$ SD)	19.2 $\pm$ 15.0	8.0 $\pm$ 13.1	19.1 $\pm$ 19.2	14.4 $\pm$ 20.4
Mean bearing ($^\circ \pm$ SD)	210 $\pm$ 55	180 $\pm$ 82	209 $\pm$ 62	146 $\pm$ 54

Elk released in Ontario, Canada, from 1998 to 2001.

Results

Between 1998 and 2001 a total of 460 elk were translocated from EINP to Ontario (Rosatte et al. 2007). Due to pre-release mortalities, 443 were released (Rosatte et al. 2007). Of these, 341 animals met our criteria for inclusion in this study (Table 3). On average, animals were located 32.2 ± 11.3 times (SD) using standard ground and aerial telemetry protocols.

Dispersal within/Between Release Areas

Mean dispersal distance ($F_{[3,340]} = 3.018$, $p = 0.03$) and dispersal direction ($F_{[3,337]} = 33.439$, $p < 0.001$) differed between the four release areas (Table 1). Elk in BNH dispersed farther than elk in LHNS ($p = 0.043$). Conversely, elk in LOW and NFR dispersed similar distances ($p = 0.461$). Dispersal directions within each release area showed a lack of uniformity (concentration: 0.534–0.739), suggesting that elk dispersed in similar directions within each release area. Elk in NFR dispersed in a different direction than elk in LOW ($F_{[1,227]} = 76.505$, $p < 0.001$) and LHNS ($F_{[1,218]} = 48.441$, $p < 0.001$). Elk in BNH also dispersed in a different direction ($F_{[1,119]} = 27.714$, $p = 0.037$) than elk in LOW and LHNS ($F_{[1,110]} = 15.912$, $p < 0.001$).

Dispersal by Sex and Age Class

Mean dispersal distance differed between sex and age classes ($F_{[3,340]} = 4.591$, $p = 0.004$). Male elk released as calves remained closer to the release site (Table 2) than adult cows ($p = 0.003$) and adult bulls ($p = 0.015$). Female calves did not move significantly farther than any other sex/age class. We expected calves to remain in close proximity to other elk and to therefore disperse in a similar direction. However, mean dispersal bearing also differed between sex/age groups ($F_{[3,337]} = 13.49$, $P < 0.001$). Female calves dispersed in a more southeasterly direction than adult males ($F_{[1,104]} = 32.263$, $P = 0.002$), adult females ($F_{[1,237]} = 37.0$, $P < 0.001$), and male calves ($F_{[1,82]} = 4.194$, $p = 0.044$).

Table 1. Mean dispersal distance and direction of elk.

	Mean Distance (km)	SD	Mean Direction ($^\circ$)	SD
BNH	22.2	18.9	183	62
LOW	19.9	30.8	237	46
LHNS	13.0	8.5	227	49
NFR	16.0	13.0	165	64

Elk released into four release areas in Ontario from 1998 to 2004.

Table 3. Demographic and dispersal characteristics of elk (*Cervus elaphus*).

Release Group	Holding Period (days)	Release Date	n	Adult Males	Adult Females	Male Calves	Female Calves	Mean Distance (km)	SD	Mean Bearing (degrees)	SD
NFR 1998	27	March	33	11	22			17.5	11.3	183	35
NFR 1999	4*	Jan 15	56	10	46			22.6	14.8	208	44
NFR 2000	43	Mar 30	36		17	7	12	7.3	7.3	141	186
NFR 2001	59	Apr 14	21	1	8	4	8	7.6	2.6	100	21
NFR 2003	44	N/A	12	1	9	1	1	22.2	10	216	72
LOW 2000	11*	Jan 29 (cows) Feb 19 (bulls)	29	1	19	6	3	26	38.3	248	54
LOW 2001	17	Mar 9	42	11	21	6	4	15.7	23.9	227	38
BNH 2001	90	Jan 9	50	14	19	6	11	22.2	18.9	196	64
LHNS 2001	112	Apr 8	42	8	24	8	2	14.6	7.6	247	21
LHNS 2004	69	Apr 8	20	5	10	2	3	9.5	9.2	167	47

Elk (*Cervus elaphus*) groups reintroduced into Ontario, Canada from 1998 to 2004.

* Denotes semi-soft release.

Influence of Holding Period Length

Data were separated into 10 release groups, each of which was held for a different length of time prior to release. Results indicated that both mean distance ($F_{[9,340]} = 4.341$, $p < 0.001$) and mean bearing ($F_{[9,331]} = 49.362$, $p < 0.001$) differed between release groups. As predicted, groups NFR 1999 and LOW 2000 (held 4 and 11 days, respectively) dispersed the furthest and clustered together in posthoc comparisons of dispersal distance. Elk from cohorts NFR 2000, NFR 2001, and LHNS 2004 were held between 43 and 69 days, and remained in closer proximity to the release site than elk from cohorts NFR 1999 and LOW 2000, where elk were held for 4 and 11 days, respectively (Table 3). We expected all release groups that experienced a soft release (17–112 days in this study) to group together in posthoc tests of dispersal distance. However, contrary to our predictions, elk from LHNS 2001 and NFR 2003 did not remain as close to the release site as other groups.

Analysis of mean dispersal bearing revealed several significant differences between release groups ($F_{[9,331]} = 49.362$, $p < 0.001$). Since all groups in a release area were released from the same facility (except LHNS 2004; see Discussion), it can be assumed that habitat characteristics were very similar (if not identical). Therefore, any differences in dispersal characteristics between groups released at the same site would not result from exposure to differing habitat. NFR 2001 dispersed in an easterly direction ($98 \pm 20^\circ$), and LOW 2000 dispersed at a bearing of $254 \pm 50^\circ$. All other release cohorts exhibited a mean dispersal bearing between 100° and 248° . We found much variability in dispersal direction within the NFR release area. Differences in dispersal bearing were found between NFR 1999 (4 days) and both NFR 2000 (43 days; $F_{[1,90]} = 157.623$, $p < 0.001$) & 2001 (59 days; $F_{[1,75]} = 128.189$, $p < 0.001$). There was also a difference between NFR 1998 (27 d.) and NFR 2001 (59 d.) ($F_{[1,52]} = 99.773$, $p < 0.001$). In LHNS, dispersal bearings were significantly different between the 2001 ($248 \pm 20^\circ$) and 2004 ($160 \pm 45^\circ$) release groups ($p = 0.003$); however, due to logistical difficulties, LHNS

2004 was released from a different holding facility approximately 55 km northwest of the previous facility. No differences were discovered between LOW 2000 ($254 \pm 50^\circ$) and LOW 2001 ($228 \pm 37^\circ$) ($p = 0.975$).

Concentration of dispersal directions varied between 0.529 and 0.939 for all release groups. Several release groups showed a high degree of concentration: LHNS 2001 (0.939), NFR 1998 (0.834), NFR 2000 (0.908), NFR 2001 (0.942), and LOW 2001 (0.809). All release groups showed a significant departure from uniform dispersal, which was not surprising considering the gregarious nature of elk.

Discussion

Elk movements are often random as animals move in relation to distribution of important resources, and agricultural, horticultural, and recreational practices (Van Dyke et al. 1998). Movement decisions also must meet forage requirements under varying levels of predation risk and maintain sociality (Anderson et al. 2005a). Considering the influence of postrelease dispersal on future demographic processes, knowledge of movement patterns is essential to understanding the establishment and colonization of a population (Machtans et al. 1996; Dieckmann et al. 1999; Dzialak et al. 2005).

Dispersal within/Between Release Areas

The two southernmost elk release areas (NFR and BNH) grouped together in posthoc tests of dispersal direction. Both release areas are dominated by mixed deciduous forests, with little or no coniferous forest patches. Human population densities and climate are similar as well. There was a significant difference in direction between NFR and both LHNS and LOW. Elk in NFR and BNH dispersed in a southerly direction while elk in LOW and LHNS dispersed more to the southwest.

The authors cannot rule out the influence of other factors with respect to dispersal. Dispersal of elk in LHNS may have

been influenced by the presence of a hydro-electric corridor in close proximity to the release site. Hydro-electric corridors are generally long and wide, contain low-lying vegetation, and are often restricted to off-road vehicles—properties that make them extremely inviting to elk. Unfortunately, the majority of elk in LHNS used the hydro-line as a unidirectional travel corridor and subsequently established themselves in the surrounding agricultural areas, where they remained there for over 3 years. It is also important to note that the two LHNS release groups (2001 and 2004) were released from different release sites. In the winter of 2004, nuisance elk from the original LHNS release were captured via helicopter and translocated to a release site approximately 55 km from the original release site. As a result, the LHNS 2004 release group consisted of several animals that had been translocated and released previously (10 of 18 elk; 55%)—a fact which may have impacted dispersal behavior.

Dispersal of elk may have been impacted by large-scale differences in habitat. There is a marked contrast between habitats in the northern and southern portions of the LHNS and LOW release areas. In LHNS, the southern landscape is dotted with large tracts of land that have been converted to agricultural uses, as well as several large patches of regenerating forests. To the north, the terrain becomes extremely rocky, and is much higher in elevation. Elk have been shown to prefer topographic zones of low elevation (McCorquodale et al. 1988). For this reason, during the release site selection phase, it was presumed that elk would not disperse to the north for this reason, even though the level of anthropogenic disturbance is extremely low.

The Lake of the Woods release area is unique in that it occurs in a transition zone between the eastern deciduous and boreal forest regions. Elk tended to disperse to the south, away from the coniferous forests typical of the boreal forest region. The southern portion of the study area is largely agricultural, providing large openings and ample forage. It also receives significantly less snowfall than the northern portion, with spring green-up occurring approximately 1 month earlier (Anonymous 2003). The dispersal and subsequent establishment of elk in the south may, therefore, be related to different forage availability resulting from local climate variation.

The presence of conspecifics has been proposed as an influential factor in dispersal behavior (Sarrazin & Legendre 2000; Le Gouar et al. 2008). There are approximately six elk farms located in the southern portion of the study area. Presumably, these elk farms may have drawn animals south in search of social relationships, particularly during the rut. The same is true of LHNS where there is a single elk farm approximately 20 km to the south (Ryckman 2006), and BNH where elk traveled to an elk farm 100 km to the south, near Peterborough, Ontario (Rosatte, personal communication).

Dispersal by Sex and Age Class

Information remains scant regarding the effects of sex and age composition on dispersal, survival, and overall sustainability of reintroduced ungulate populations (Komers & Curman 2000;

Bar-David et al. 2005). The sex and age ratio of a reintroduced population is an important consideration, especially in ungulate populations where female estrus is induced by males. The distribution of females will undoubtedly impact the distribution of males on the landscape (Haydon et al. 2007). On average, calves remained closer to the release site than adults. Furthermore, calves in this study dispersed in a significantly different direction than adults. This result was unexpected, as our analysis included data up to 2 years postrelease, and calves typically remain with adult females for 1.5–2 years before dispersing (Geist 1982). This may have resulted from individual calves being translocated without their mothers or matrilineal group. Larkin et al. (2004) noticed a similar trend in elk populations released into three areas in Kentucky. They concluded that, in situations where elk are required to remain within a small area surrounding the release site, it may be advantageous to release a greater proportion of calves. Young animals often suffer higher natural mortality than adults, an effect which is offset by resiliency to the effects of capture. While adults experience higher survival in natural populations, they tend to be adversely affected by the capture process (Sarrazin & Legendre 2000). Unfortunately, releasing a higher proportion of calves could result in delayed population growth due to the amount of time (~ 2 years) required for young elk to achieve sexual maturity (Larkin et al. 2004). A key goal of reintroduction programs is to increase population size as quickly as possible, thereby limiting the problems associated with small populations (e.g., genetic depression, effects of predation, catastrophic events, etc.). Sarrazin & Legendre (2000) modeled the demographic consequences of releasing adult versus juvenile animals on reintroduction success and concluded that it is more efficient to release adults than juveniles, in part because one of the criteria of reintroduction success is the production of a wild-born generation as soon as possible. If elk were released as calves or yearlings, this process would be delayed until the onset of sexual maturity.

Elk typically disperse from matrilineal social groups in the spring of their second year. Dispersal in elk, as in other polygynous species, is expected to be male-biased due to variability in reproductive success and foraging behavior (Geist 1982; Petersburg et al. 2000). Three hypotheses have been proposed to explain this behavior: mate competition, resource competition, and inbreeding avoidance (Petersburg et al. 2000). As predicted, mean dispersal distances were significantly greater for adult male elk than calves of either sex. However, no significant difference was discovered between adults of either sex, even though the greatest distance traveled by an elk in each release area was performed by an adult female. Likewise, calves remained in closer proximity to release sites in BNH in 2000/2001 (Yott 2005) and in LOW in 2000 (McIntosh 2003). Due to the processing procedures at EINP, it is difficult to determine if calves were captured and released with their mothers. However, genetic analysis of reintroduced elk is in process. Moran (1973) discovered a similar trend in Michigan's Rocky Mountain elk population, where greatest movements (9–19 km) were performed by four adult cows. However, the number of elk collared was skewed

in favor of females, suggesting that this estimate may be biased. Based on field observations, it was concluded that adult males travel more extensively than females and are more often involved in long-distance movements (Moran 1973). Elk released in LHNS did not disperse as far as those released in BNH, despite being held for a longer period. Haydon et al. (2007) studied the BNH elk population and found that solitary elk dispersed farther than elk in groups. They also found that males dispersed farther than females during the second and third years postrelease, with dispersal becoming similar for both sexes by the fourth year postrelease.

Maximum dispersal distances ranged from 28 to 42 km, although no significant differences were found between release areas (Ryckman 2006). These distances are drastically different from those seen in Kentucky's reintroduced elk population, where elk released in three different areas dispersed 90, 124, and 152 km after 1 year (Larkin et al. 2004). Unfortunately, without performing fine-scale habitat analyses in each release area, it is impossible to determine the precise influence of habitat on the dispersal characteristics of Ontario's elk populations. Yott (2005) studied elk that were hard-released in BNH and found that many elk had dispersed >100 km. Haydon et al. (2007) found similar maximum dispersal distances in BNH, where elk had dispersed 5–97 km from the release site after 1,000 days to cover an area >10,000 km².

Influence of Holding Period Length

The virtues of a soft release are often insinuated (Jeffries et al. 1986; Gogan & Barrett 1988; Stanley Price 1989; Tuberville et al. 2005), although few studies have directly tested the effect of release method (Bright & Morris 1994) on dispersal characteristics. In fact, Gogan and Barrett (1988) concluded that degrees of site fidelity and social cohesion seen in Tule elk (*Cervus elaphus nannodes*) reintroduced in California were the result of holding animals in small pens for 3 to 6 months prior to release. Furthermore, Haydon et al. (2007) studied two different elk releases in BNH and found that release method had no effect on postrelease survival. Cromwell et al. (1999) studied the effect of relocation on white-tailed deer (*Odocoileus virginianus*) in South Carolina, where 50% of relocated deer dispersed from the release site. While it would have been beneficial to include a soft release group for comparative purposes, these results suggest that relocation may be sufficient to induce dispersal behavior in deer. Elk in the present study experienced holding periods between 4 and 112 days. As predicted, cohorts NFR 1999 and LOW 2000 dispersed a greater mean distance than other cohorts, which may have been due to the short pre-release conditioning periods (4 and 11 days, respectively). An extended conditioning period is expected to be beneficial by allowing translocated animals to become accustomed to a new area, and giving animals an opportunity to recuperate from the stress of capture and transportation.

Since elk in cohort LHNS 2001 experienced the longest pre-release conditioning period, we expected them to remain closer to the release site than elk in any other cohort. However,

mean dispersal distance for this cohort was not significantly different from any other group. Conversely, we expected elk from LOW 2001 to display similar dispersal patterns as NFR 1999 and LOW 2000 due to similarities in holding period length. In fact, LHNS 2001 and LOW 2001 dispersed approximately the same distance (14.6 and 15.7 km, respectively), even though their holding periods were considerably different (112 and 17 days, respectively). Again, LOW 2001 and LHNS 2001 did not conform to our predictions, as they dispersed at a moderate distance compared to other cohorts. These results suggest that mean distances in Ontario's elk populations cannot entirely be explained by differences in holding period length and that there may be a plateau above which it is no longer beneficial to confine elk.

Our analysis did not include information regarding time of year at release. Elk were intended to be released during the spring months to avoid snow cover, and to coincide with the growth of new vegetation. However, in cases where cohorts experienced a semi-soft release (LOW 2000 and 2001, NFR 1999), this release typically occurred prior to spring. Thus, elk may have needed to disperse greater distances in order to find suitable forage and cover habitats, thereby skewing the results. However, this is not likely the case, as release areas were selected as having the most suitable elk habitat in Ontario. Furthermore, it has been suggested that hard releases may result in animals ignoring supplementary food, which would further compound the problem (Bright & Morris 1994).

Conclusions

Demographic and spatial characteristics of reintroduced populations are almost certainly affected by the reintroduction process, and researchers must therefore take this into account when planning future reintroduction programs. The impacts of postrelease dispersal on subsequent vital rates (reproduction, survival, home range establishment, etc.) are important considerations when planning future reintroductions. Managers must consider all aspects of a species' ecology when initiating similar programs, including the species' biology, habitat preferences, and ultimately the goals of the program. In our study, calves remained closer to release sites than adults, suggesting that calves may be less likely to make exploratory movements. Due to the processing procedures at Elk Island National Park, it is difficult to determine if calves were captured, transported, and released with their mothers. This may have resulted in the lower dispersal by calves. However, genetic analysis is underway to characterize relationships between translocated elk. Also, calves generally dispersed in a different direction than adults, which is perhaps an evolutionary strategy to avoid inbreeding.

The vast majority of dispersal at all sites was in a southerly direction, towards the core of elk's historic range in Ontario. This phenomenon may have been influenced by preferred habitat to the south of most release sites. Also, climate also improves as one travels south, with a decrease in mean snow depth, number of snow days, and higher mean temperatures. Furthermore, the prevailing winds in Ontario

(south-westerly/westerly in the summer) may have contributed to elk dispersal direction, as elk face the wind to detect predators. This phenomenon should be accounted for in future elk reintroductions into Ontario. It would be advantageous to release animals at the northern periphery of a designated management area, thus allowing them to disperse to the south into areas of preferred habitat and climate.

Managers must account for all possible determinants of dispersal when considering ungulate reintroductions. Finally, dispersal characteristics were affected by holding period length, with significantly lower dispersal distances seen in elk that were held for moderate periods. Longer pre-release conditioning periods may allow elk to acclimate to their new environment prior to being forced to fend for themselves, and may also strengthen social bonds between unrelated individuals. Clearly, a longer pre-release conditioning period could be advantageous when managers want animals to remain within a predefined management area.

Implications for Practice

- Considering the number of reintroductions that take place, managers must consider the influence of demography when releasing groups of animals into areas of former occupation.
- Increased holding time prior to release should be considered as an option in situations where managers want to confine animals to a particular area.
- A detailed understanding of all factors that influence postrelease dispersal is of vital importance when planning an ungulate reintroduction program.

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